

AZMAIN IQTIDAR ARNOB

Software Engineer — Python Backend & Applied AI

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SUMMARY

Final-year Computer Science & Engineering student at North South University, graduating October 2026, focused on backend software development in Python. Built role-based web applications with Flask and MongoDB, real-time desktop systems with Electron, WebRTC and Socket.IO, and retrieval-augmented AI assistants. Grounded in data structures, algorithms, object-oriented programming and database design. Seeking an entry-level Software Engineer role in Dhaka.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript, SQL

Backend & Web: Flask, Django, REST API design, Socket.IO, Jinja2, Tailwind CSS, HTML/CSS, Electron

Databases: MongoDB, MySQL, schema design, query optimisation, role-based access control

AI & Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, OpenCV, RAG systems, vector databases, LLM integration

Cloud & Tools: AWS fundamentals, Git & GitHub, Linux, Google Colab, Agile & pull-request workflow

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering

PROJECTS

LabDesk — LAN Classroom & Lab Management Platform | Electron, Socket.IO, WebRTC, SQLite *Senior Design Capstone, CSE 499*

- Developed both frontend and backend of a Windows lab-management suite, pairing an Electron client with an embedded Express and Socket.IO server over SQLite to give instructors live monitoring and remote control of student workstations from a single console.
- Implemented peer-to-peer screen streaming over WebRTC, removing the server as a relay bottleneck during concurrent classroom broadcasts.
- Built an exam-lockdown mode enforcing URL blocking, USB and Bluetooth device locks and application whitelisting, backed by a SYSTEM-level scheduled task guaranteeing safe-state restoration; currently integrating Agentic AI to automate classroom workflows.

HAMS — Hospital Appointment Management System | Flask, MongoDB, Tailwind CSS *CSE 411, Advanced Databases*

- Built a role-based web application serving patient, doctor and administrator workflows, securing every protected route with bcrypt-hashed credentials and session-based access control across three distinct roles.
- Designed a booking engine validating each request against doctor availability and rejecting double-bookings server-side, supporting a full pending, confirmed, completed and cancelled appointment lifecycle with digital prescriptions.
- Modelled users, appointments and prescriptions as MongoDB documents with filtered queries and status aggregation driving the administrator dashboard.

Travelmate BD — Retrieval-Augmented Tourism Assistant | Python, ChromaDB, Gradio *CSE 299 Junior Design, 4-person team*

- Built a bilingual Bangla and English travel assistant combining a locally hosted LLM with document-grounded question answering over uploaded PDF and DOCX files using vector retrieval on sentence-transformer embeddings.
- Added an algorithmic tour-budget estimator and a psutil-based performance monitor tracking runtime memory consumption and response latency.

VisionGuard — Multi-Label Retinal Disease Classification | PyTorch, TensorFlow, scikit-learn *CSE 445 & CSE 465*

- Trained and compared Random Forest, XGBoost on HOG features, a custom CNN and a fine-tuned ResNet-50 across 45 disease labels on the RFMiD fundus dataset under severe class imbalance.
- Fine-tuned ResNet-50 outperformed the from-scratch CNN baseline by roughly five times on macro-F1; added Grad-CAM explainability plus training-time and memory profiling.

UNDERGRADUATE THESIS

Instance-Aware Object Removal & Background Reconstruction | YOLOv8, SAM, LaMa *CSE 498R, Directed Research*

- Built a hybrid inpainting pipeline performing progressive instance-wise removal of people from crowded real-world scenes, and introduced a residual-aware evaluation metric using object re-detection to measure removal completeness.

EDUCATION

North South University — BSc in Computer Science & Engineering, Dhaka *Expected October 2026*

CGPA: 3.11 / 4.00

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Advanced Database Systems, Software Engineering, Machine Learning, Pattern Recognition & Neural Networks, Operating Systems, Computer Networks

TRAINING & LANGUAGES

AWS Cloud Practitioner Essentials — AWS Skill Builder, 2026; currently preparing for the AWS Certified Cloud Practitioner exam.

Languages: Bengali (native) · English (fluent) · Hindi (conversational)